

Newton's laws

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

QUESTION		ANSWER	✓
1	State Newton's first law.	If the resultant force is zero, a stationary object stays still and a moving object keeps the same velocity.	☐
2	An object travels at constant velocity. Is there a resultant force on it?	No	☐
3	Define inertia.	The tendency of an object to continue in its state of rest or of uniform motion.	☐
4	State Newton's second law.	Acceleration is proportional to the resultant force and inversely proportional to the mass ($F = m a$).	☐
5	Describe how the acceleration of an object depends on the resultant force.	Acceleration is directly proportional to the resultant force.	☐
6	What is inertial mass?	A measure of how difficult it is to change the velocity of an object.	☐
7	An object is changing velocity. Is there a resultant force on it?	Yes	☐
8	State Newton's third law.	When two objects interact, the forces they exert on each other are equal and opposite.	☐
9	State the equation linking resultant force, mass and acceleration.	resultant force = mass $\times$ acceleration ($F = m a$)	☐
10	For a constant force, state how acceleration depends on mass.	Acceleration is inversely proportional to mass.	☐
11	A car travels at a steady speed along a straight road. Compare the driving force and the resistive forces.	They are balanced (equal and opposite).	☐
12	State the meaning of this symbol: $\sim$	Approximately	☐

QUESTION		ANSWER	✓
13	Give an example of Newton's third law.	You push on a wall; the wall pushes back on you with an equal and opposite force.	☐